Lecture 8: Disability Insurance

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Disability Insurance

- Golosov-Tsyvinski JPE 2006
- Disability insurance is an important social insurance program
- Theoretical study of the design of an optimal welfare program
- US (2000): At the time
  - Second largest welfare program in the U.S.: 55 billion dollars
  - 6 million beneficiaries
  - Three times as large as unemployment insurance
- Example: 20% of working age males on disability in Netherlands (1994)

Motivation

- Legal definition: "inability to engage in substantial gainful activity"
- Disability is difficult to determine. In practice, disability determination is very subjective
- Even medical criteria are difficult to verify (back pain, chronic exhaustion, stress, etc.)
- Both medical and situational factors have to be considered
- Multiple impairments

Setup

- Diamond and Mirrlees (1978, 1986)
- Each agent lives for N periods
- 2 types of agents in each period: able and disabled
- disability is unobservable
- interest rate R, wage w fixed
- discount factor β
- Separable utility $U(c,l) = u(c) - h(l)$

How G-T define disability

- In period t: able ($\theta_t > 0$) or disabled ($\theta_t = 0$)
- Effective labor $y = \theta l$
- Once disabled, stays disabled forever.
- Information: Disability is unobservable
  - θ and l are not observable
  - y is observable
  - $y > 0$ must be able
  - $y = 0$ either disabled or able claiming disability

Two-period example

- in period 1: able ($\theta = 1$)
- In period 2:
  - able ($\theta = 1$) with probability p
  - disabled able ($\theta = 0$) with probability $1 - p$
- Social planner's problem

$$\max_{c,y} u(c_1) - h(y_1) + \beta p[u(c_a) - h(y_a)] + \beta(1-p)u(c_d)$$

$$u(c_a) - h(y_a) \geq u(c_d) \qquad \text{(IC)}$$

$$c_1 + \frac{pc_a}{1+R} + \frac{(1-p)c_d}{1+R} \leq wy_1 + \frac{pwy_a}{1+R} \qquad \text{(Feasibility)}$$

- If disability were observable
  - Everybody would be fully insured: consumption of able and disabled equal
  - No savings distortion $u'(c_1) = \beta(1+R)u'(c_2)$
- If disability imperfectly observable
  - Inverse Euler Equation: Rogerson (1995)

$$\frac{1}{u'(c_1)} = \frac{1}{\beta(1+R)}[\frac{p}{u'(c_a)} + \frac{1-p}{u'(c_d)}]$$

  implies that savings should be distorted

$$u'(c_1) < \beta(1+R)[pu'(c_a) + (1-p)u'(c_d)]$$

Main purpose of paper: Implement the optimum

- Implementation: Given taxes and transfers an agent chooses the optimal allocation.
- **Key difficulty:** how to translate results about wedges in the results about optimal taxes.
- **Approach:** Implement optimal allocation using simple asset-tested disability transfers
- In practice, asset-testing not in SSDI, but in other welfare programs.

Linear Tax can not implement the optimum

- One might guess that the following tax system will implement the optimum:
- Savings tax τ to satisfy intertemporal FOC

$$u'(c_1^*) = \beta(1 + (1 - \tau)R)[pu'(c_a^*) + (1 - p)u'(c_d^*)]$$

- lump sum taxes to satisfy budget constraints

  $$c_1^* + k^* = wy_1^* + T_a, \qquad c_a^* = [1 + R(1 - \tau)]k^* + T_a, \qquad c_d^* = [1 + R(1 - \tau)]k^* + T_d$$

  for some level of capital k^*

- But this is incorrect ... **Without asset-testing a joint deviation: oversave and claim disability can be profitable.**
- Linear savings tax ensures that:
  - Given truth-telling, agent chooses correct savings
  - Given correct savings, agent chooses to tell the truth
- Does not rule out joint deviation: oversave and claim disability

- SP problem: IC binds $u(c_a^*) - h(y_a^*) = u(c_d^*)$
- $(c_1^*, y_1^*, c_a^*, y_a^*, c_d^*)$ gives the same utility as $(c_1^*, y_1^*, c_d^*, 0, c_d^*)$
- Can an agent deviate and get a better utility than $(c_1^*, y_1^*, c_d^*, 0, c_d^*)$?
- An agent planning to claim disability in the second period, regardless of his true type, solves the following problem.

$$\max u(c_1) - h(y_1) + \beta u(c_2)$$
$$c_1 + k_2 = wy_1 + T_a$$
$$c_2 = (1 + (1-\tau)R)k_2 + T_d$$

- $(c_1^*, y_1^*, c_d^*, 0, c_d^*)$ is feasible for this problem
- But it is not optimal
- FOC of problem evaluated at optimal allocation

$$u'(c_1^*) = \beta(1 + (1-\tau)R)u'(c_d^*)$$

- But from definition of tax τ

$$u'(c_1^*) = \beta(1 + (1-\tau)R)[pu'(c_a^*) + (1-p)u'(c_d^*)]$$

- Both can hold iff $c_a^* = c_d^*$ which can't be true if IC binds. **Tax is not sufficient!**

Asset Testing

- Golosov-Tsyvinski construct asset-tested disability transfers that implement the optimal allocations
- Period 1: lump-sum tax equal to T_a
- Period 2: asset-tested disability $\{(T_a, T_d), k_2^*\}$
  - T_d if $k_2 \leq k_2^*$ and $y = 0$ (receive a disability transfer)
  - T_a otherwise (treated as able)
- References on Tax Implementation:
  - Werning, Ivan (2012) "Nonlinear Capital Taxation" mimeo MIT
  - S. Albanesi and C. Sleet, "Dynamic Optimal Taxation with Private Information," Review of Economic Studies 73 (2006), 1-30.

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